

The Illusion of Parity: Human Cultural Adequacy Reveals Shared Whitewashing in Dialectal Translation

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Abstract

Large language models (LLMs) achieve competitive scores on standard machine translation (MT) benchmarks, yet may *over-normalise* minority dialects by flattening register, pragmatics, and local cultural meaning—a phenomenon we term *cultural whitewashing*. We study Sichuanese→English translation on a 200-sentence keyword-stratified probe set derived from WenetSpeech-Chuan. Two contemporary Chinese-strong systems—GLM-4-Plus and DeepSeek-V4-Flash¹—yield nearly identical system-level COMET scores (70.76 vs. 70.87; paired bootstrap $p = 0.836$), suggesting parity at the corpus level. Human *cultural adequacy* judgments tell a different story: only 14% of sentences receive culturally adequate translations from *both* models, whereas 28% fail from *both* (*consensus whitewashing*). Automatic metrics partially separate successful from failed cases (mean COMET gap ≈ 12 points between *tie-good* and *tie-bad* labels), yet culturally failed outputs still score ≈ 67 COMET—only ≈ 4 points below the corpus mean—and human-COMET winner agreement on contested sentences is just 46.6%. We release our probe construction pipeline and present a lightweight 2×2 human protocol ($\kappa = 0.71$) for variation-rich NLG evaluation.

1 Introduction

Dialectal and low-resource speech communities increasingly interact with LLM-based translation and generation systems trained predominantly on standard-language data (Hendy et al., 2023). When such systems map regional varieties into globally dominant languages, they often produce fluent outputs that nonetheless erase pragmatic force, social deixis, and culture-specific lexical se-

mantics (Freitag et al., 2021). We refer to this systematic loss of culturally marked meaning as **cultural whitewashing**.

Evaluating whitewashing requires more than corpus-level string overlap. On our Sichuanese probe set, GLM-4-Plus and DeepSeek-V4-Flash—representative strong LLMs accessible via public APIs—achieve statistically indistinguishable COMET and BLEU (Table 2), creating an **illusion of parity (RQ1)**. Do such aggregate scores hide localized pragmatic failure (**RQ2**)? When both models fail, do they fail in homogeneous ways that automatic metrics under-penalize (**RQ3**)?

We address these questions with a stratified evaluation framework combining: (1) keyword-based sampling over WenetSpeech-Chuan transcriptions to enrich content-bearing dialect terms; (2) system-level and keyword-stratified automatic metrics (BLEU, chrF, COMET); (3) a **2×2 human cultural adequacy** scheme that labels each sentence according to whether *each* model successfully transmits source cultural/pragmatic meaning—not merely which translation reads more fluently.

Headline findings. Only 28/200 sentences (14.0%) are *tie-good* (both models adequate); 56/200 (28.0%) are *tie-bad* (consensus failure). DeepSeek-V4-Flash passes human cultural adequacy on 52.0% of sentences vs. 34.0% for GLM-4-Plus, yet system COMET differs by only 0.11 points. COMET separates *tie-good* from *tie-bad* sentences by ≈ 12 points, but *tie-bad* outputs still average 66.9 COMET vs. a global mean of 70.8. On 116 sentences where exactly one model is adequate (*g* or *d*), COMET’s preferred model agrees with humans only 46.6% of the time. Five curated cases (Table 6) show COMET rewarding whitewashed paraphrases.

Contributions. (1) A reproducible keyword-stratified Sichuanese probe set from WSC-Train. (2) A lightweight human protocol for cultural ad-

¹We invoke the public DeepSeek Chat API (deepseek-chat); our provider dashboard reports usage under DeepSeek-V4-Flash.

equacy with substantial inter-annotator agreement ($\kappa = 0.71$). (3) Quantitative evidence of metric–human divergence and consensus whitewashing on two LLMs. (4) A qualitative typology motivating culturally resilient evaluation for personalised and variation-rich NLG.

2 Related Work

Dialectal resources and MT. WenetSpeech-Chuan (WSC-Train) provides large-scale Sichuanese speech transcriptions derived from in-the-wild data (Anonymous, 2025). Dialect-aware MT remains challenging: models trained on standard Mandarin or English often normalize regional forms toward mainstream lexicon and syntax.

LLM translation and metric validation. LLMs translate competitively on high-resource pairs (Hendy et al., 2023), but reference-based metrics such as BLEU (Post, 2018), chrF (Popović, 2015), and COMET (Rei et al., 2022) assume that human references encode the semantic targets worth preserving. When references are themselves culturally marked—as in our human-written Sichuanese→English gold—metrics may still reward generic fluent alternatives that align partially with reference wording while missing source pragmatics (Freitag et al., 2021).

Human evaluation beyond fluency. Expert and error-span annotations reveal systematic gaps between metric rankings and human judgments (Freitag et al., 2021). Our work extends this line to *cultural adequacy* on dialectal input, using a four-way label scheme designed to detect *shared* failure rather than pairwise preference alone.

3 Methodology

3.1 Probe Set Construction

Source corpus. We download `wsc_metadata.jsonl` from HuggingFace dataset ASLP-lab/WSC-Train (Anonymous, 2025) and retain utterances with ASR confidence ≥ 0.9 , yielding 2,598,002 sentences.

Keyword stratification. We define 45 keyword patterns grouped into (i) *content-bearing* dialect terms (e.g., 老子, 安逸, 瓜娃子, 咋个), (ii) *modal particles* (嘛, 哈, 哦), and (iii) *long-tail cultural tokens* (e.g., mahjong term 九通, 扯把子). High false-positive single-character patterns use context

constraints; filler tokens and ASCII fragments are removed. Sentences must contain at least one content keyword, have length ≤ 80 characters, and are selected via weighted sampling ($n = 200$, seed 42): content terms weight 1.0, particles 0.15, with upweights for rare insult/register terms.

Reference translations. Two bilingual annotators independently authored culturally faithful English references for all 200 sentences. References are used *only* for automatic metric computation; models never see them at inference.

3.2 Translation Systems

Both models receive an identical naive prompt: “*You are an expert translator.*” (system) and “*请将以下四川话翻译为地道的英文: {text}*” (user). We call GLM-4-Plus via Zhipu API (`glm-4-plus`) and DeepSeek-V4-Flash via DeepSeek Chat API (`deepseek-chat`). We compare exactly these two systems; results do **not** generalize to broader open-vs. closed-source categories.

3.3 Automatic Metrics

We compute corpus-level BLEU and chrF (sacreBLEU) and sentence-level COMET (Unbabel/wmt22-comet-da, reported $\times 100$) against human references. System-level differences are tested with paired bootstrap resampling ($B = 2000$; Koehn, 2004). We additionally report keyword-stratified chrF/COMET (minimum $n = 5$ per keyword in main tables) and flag *refusal* outputs—fluent meta-responses instead of translations—via regex patterns (GLM-4-Plus: 1/200; DeepSeek-V4-Flash: 0/200).

3.4 Human Cultural Adequacy Protocol

Task. Annotators view the Sichuanese source and two blinded English translations (model identity hidden). For each sentence they assign one of four **consensus labels** based on whether each translation *successfully transmits the cultural/pragmatic meaning* of the source (not general fluency alone):

Annotators and agreement. Two bilingual annotators independently labeled all 200 sentences, then discussed disagreements to reach consensus gold labels. Pre-consensus Cohen’s $\kappa = 0.71$ indicates substantial agreement (Cohen, 1960). Guidelines emphasize that 安逸 encodes Sichuanese pleasure/satisfaction (not merely “comfortable”),

Label	GLM	DS	Meaning
<i>tie-good</i>			Both culturally adequate
<i>g</i>		×	Only GLM-4-Plus adequate
<i>d</i>	×		Only DeepSeek-V4-Flash adequate
<i>tie-bad</i>	×	×	Consensus whitewashing

Table 1: Human cultural adequacy labels ($n = 200$). DS = DeepSeek-V4-Flash.

Model	BLEU	chrF	COMET	Cultural pass
GLM-4-Plus	16.12	39.75	70.76	34.0%
DeepSeek-V4-Flash	16.32	41.73	70.87	52.0%

Bootstrap p : COMET 0.836; BLEU 0.861; chrF 0.006.
Cultural pass = (*tie-good* + model-exclusive pass) / 200.

Table 2: System-level automatic scores and human cultural pass rates on $n = 200$ sentences.

Keyword	n	GLM-C	DS-C	GLM-F	DS-F
老子	18	66.78	66.72	34.36	37.53
安逸	42	68.34	69.08	34.02	36.59
瓜娃子	5	76.94	69.82	46.14	41.89
咋个	11	71.98	75.26	46.16	51.49
啥子	44	70.71	71.40	40.29	41.70

Table 3: Keyword-stratified scores ($n \geq 5$). C = COMET ($\times 100$); F = chrF. DS = DeepSeek-V4-Flash.

Label	Count	Percent
<i>tie-good</i> (both adequate)	28	14.0%
<i>g</i> (GLM only)	40	20.0%
<i>d</i> (DeepSeek only)	76	38.0%
<i>tie-bad</i> (consensus failure)	56	28.0%

Table 4: Human cultural adequacy gold distribution ($n = 200$).

老子 encodes emphatic self-reference (not “father” or “Laozi”), and 瓜娃子 preserves insult register.

4 Results

4.1 System-Level Automatic Scores (RQ1)

Table 2 reports system-level scores. DeepSeek-V4-Flash leads on all three metrics, but only chrF differs significantly (+1.98, 95% CI [0.62, 3.40], $p = 0.006$). COMET (+0.11, $p = 0.836$) and BLEU (+0.20, $p = 0.861$) show no significant difference. Sentence-level oracle upper bounds (selecting the better model per sentence) reach COMET 73.35, indicating headroom neither model captures alone, yet this does not reveal *where* cultural meaning is lost.

4.2 Keyword-Stratified Valleys (RQ2)

Aggregate parity masks local deficits on content-bearing terms. Table 3 shows keyword-stratified COMET for keywords with $n \geq 5$. 老子 (emphatic self-reference) averages COMET 66.8 for both models—4+ points below the corpus mean—with a 50.0% *tie-bad* rate (9/18 sentences). 安逸 (Sichuanese pleasure) averages COMET 68.3–69.1 with chrF as low as 34.0 (GLM). High-frequency particle 啥子 ($n = 44$) yields near-parity COMET (≈ 71), illustrating that stratification separates pragmatic content words from discourse markers.

4.3 Human Gold Distribution (RQ2–3)

Table 4 summarizes gold labels. The modal class is *d* (76/200; 38.0%); DeepSeek-V4-Flash alone preserves cultural meaning. Yet 56 sentences (28.0%)

are *tie-bad*, demonstrating that whitewashing is often *consensus* rather than model-specific. Only 28 sentences (14.0%) are *tie-good*.

Automatic vs. human divergence. Table 5 compares mean automatic scores on *tie-good* vs. *tie-bad* subsets. COMET increases by ≈ 12 points for both models between these groups, confirming that metrics detect *some* cultural success signal. However, *tie-bad* sentences still average 66.2–66.9 COMET—only ≈ 4 points below the global mean—suggesting metrics *underestimate* whitewashing severity.

On 116 *g/d* sentences where humans identify a unique adequate model, COMET’s segment-level winner matches the human winner in only 54 cases (46.6%). Among *tie-bad* sentences, 50.0% still assign higher COMET to DeepSeek-V4-Flash; mean *tie-bad* DS COMET is 66.88 vs. 70.87 globally.

4.4 Reverse Misalignment Cases

Table 6 lists five cases where COMET and human judgments diverge most instructively (from `reverse_misalignment_cases.csv`). Case 1: human gold *d* (only DeepSeek adequate on 瓜娃子 insult), yet COMET favors GLM by +14.1. Case 4–5: *tie-bad* consensus failures on 咋个 and 老子 despite COMET ≈ 80 and ≈ 53 respectively—fluent normalization hides pragmatic loss.

5 Qualitative Analysis

Manual inspection of *tie-bad* and selective *g/d* cases yields three recurring whitewashing patterns:

	<i>tie-good</i>	<i>tie-bad</i>	Δ
GLM COMET	78.95	66.22	+12.72
DS COMET	78.65	66.88	+11.77
GLM chrF	47.41	35.60	+11.81
DS chrF	46.16	38.23	+7.92

Table 5: Mean automatic scores by human label subset. DS = DeepSeek-V4-Flash.

Pragmatic flattening. 老子 in imperative contexts (e.g., 给老子坐车) is rendered as polite offers (“take the bus for you”) rather than emphatic demand, losing Sichuanese self-reference force (Case 5).

Literal standardization. 安逸 maps to “comfortable” or “satisfying” instead of embodied regional pleasure (Case 2).

Long-tail cultural collapse. Domain-specific terms such as mahjong 九通 become hallucinated literal objects (“wine barrel”; Case 3), which COMET may still prefer when surface fluency aligns partially with reference wording.

6 Limitations

Our study has several bounds. (1) **Scope:** one dialect (Sichuanese), one direction (into English), $N = 200$, and keyword-biased sampling—not a random corpus sample. (2) **Models:** two Chinese-strong LLMs only; we do not claim generalizations about open- vs. closed-source ecosystems. (3) **Source noise:** WSC-Train transcriptions contain ASR artifacts; some *tie-bad* labels may conflate transcription noise with model error. (4) **Reference bias:** a single culturally marked reference style affects COMET; multi-reference or reference-free cultural metrics may behave differently. (5) **Prompt:** a single naive translation prompt may understate model capability; we leave prompt engineering for future work.

7 Conclusion

System-level COMET parity between GLM-4-Plus and DeepSeek-V4-Flash on Sichuanese→English translation is misleading. Human cultural adequacy reveals that shared whitewashing is frequent (28% *tie-bad*) and that automatic metrics—while correlating with broad success/failure—assign high scores to many culturally inadequate outputs and disagree with humans on nearly half of contested cases. We argue that variation-rich NLG evaluation

should report stratified human cultural adequacy alongside reference-based metrics, especially for personalised and dialectal generation settings.

Ethics Statement

Our probe derives from publicly released speech transcriptions. Human references and annotations were produced by the authors’ team under informed internal guidelines. Dialect examples include insult/register terms used in context for linguistic analysis; we do not endorse pejorative language.

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Omitted for review.

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#	Gold	GLM-C	DS-C	Kw.	Takeaway
1	<i>d</i>	82.9	68.8	瓜娃子	COMET prefers bland “fool”; human accepts DS insult register
2	<i>d</i>	67.1	55.9	安逸	COMET rewards “comfortable”; human accepts DS pleasure reading
3	<i>g</i>	42.0	49.1	么/九通	Mahjong culture collapsed; COMET still prefers DS
4	<i>tie-bad</i>	77.8	81.7	咋个	Both miss dialect address; COMET ≈ 80
5	<i>tie-bad</i>	51.7	53.6	老子	Both normalize self-reference; moderate COMET

Table 6: Reverse misalignment cases: COMET vs. human cultural adequacy. C = COMET ($\times 100$).

A Annotation Guidelines (Excerpt)

Annotators judge **cultural/pragmatic transmission**, not fluency alone.

安逸. *Acceptable*: conveys spontaneous Sichuanese pleasure or satisfaction in context. *Reject*: generic “comfortable” with no regional pragmatic color.

老子. *Acceptable*: emphatic first-person force appropriate to imperative/boast contexts. *Reject*: “father”, “Laozi”, or de-emphasized polite paraphrase.

瓜娃子. *Acceptable*: preserves insult/register (“fool/idiot” in contextually appropriate tone). *Reject*: neutral “silly person” that defuses pragmatic aggression.

B Reproducibility

Scripts `download_sichuanese_data.py`,
`filter_seeds.py`,
`translate_experiment.py`, and
`analyze_scores.py` reproduce the probe, translations, and scores. Data: ASLP-lab/WSC-Train;
outputs: `translation_results_200.csv`,
`translation_segment_scores.csv`.